WORKED SOLUTION

STEP I 2011, Question 8

Pure • Diophantine Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q8

The key idea

The cube function is strictly increasing, so comparing m^3 with n^3 and $(n+1)^3$ is the same as comparing m with n and $n+1$. If m lies *strictly* between two consecutive integers it cannot be an integer. That leaves only a short list of values of n to try by hand. Part (ii) uses the same squeeze on a new equation.

Solution

Throughout we use the fact that $x \mapsto x^3$ is strictly increasing on all of $\mathbb{R}$, so for real a, b,

$$a < b \iff a^3 < b^3.$$

Part (i)(a)

From $(*)$,

$$m^3 - n^3 = n^2 + 1 > 0,$$

since $n^2 \geq 0$. So $m^3 > n^3$, and therefore $m > n$.

Next, $m < n+1$ if and only if $m^3 < (n+1)^3$. Using $(*)$ and expanding,

$$\begin{aligned} m^3 < (n+1)^3 &\iff n^3 + n^2 + 1 < n^3 + 3n^2 + 3n + 1 \\ &\iff 0 < 2n^2 + 3n, \end{aligned}$$

which is the required equivalence. Every step is reversible, so the "if and only if" holds.

Now $2n^2 + 3n = n(2n+3)$. This is a quadratic with roots $n = -\frac{3}{2}$ and $n = 0$ and a positive leading coefficient, so it is positive exactly when $n < -\frac{3}{2}$ or $n > 0$. For any such n we have both $m > n$ and $m < n+1$. Hence

$$n < m < n+1 \quad \text{unless} \quad -\tfrac{3}{2} \leq n \leq 0.$$

Part (i)(b)

Suppose m and n are integers satisfying $(*)$. If $n < -\frac{3}{2}$ or $n > 0$, part (a) gives $n < m < n+1$. No integer lies strictly between the consecutive integers n and $n+1$, so this is impossible.

? **Ask yourself** Part (a) holds for all real n. Where does being an integer come in?

It comes in twice. It rules out every n outside the interval, as above, and it cuts the interval $-\frac{3}{2} \le n \le 0$ down to $n \in \{-1, 0\}$. We check both values:

- $n = 0$: $m^3 = 0 + 0 + 1 = 1$, so $m = 1$.
- $n = -1$: $m^3 = -1 + 1 + 1 = 1$, so $m = 1$.

Both give integer m. So the only integer solutions are $(m, n) = (1, 0)$ and $(1, -1)$, as required.

Part (ii)

We copy part (i). First compare p with q:

$$p^3 - q^3 = 2q^2 - 1,$$

which is positive whenever $q^2 > \frac{1}{2}$, so for every integer $q \neq 0$. For those q we have $p > q$.

Next compare p with $q+1$:

$$\begin{aligned} p^3 < (q+1)^3 &\iff q^3 + 2q^2 - 1 < q^3 + 3q^2 + 3q + 1 \\ &\iff 0 < q^2 + 3q + 2 = (q+1)(q+2). \end{aligned}$$

This holds exactly when $q > -1$ or $q < -2$.

Put the two together. If q is an integer with $q \neq 0$, $q \neq -1$ and $q \neq -2$, then $q < p < q+1$, so p cannot be an integer. (If $q \ge 1$ or $q \le -3$, both conditions hold.) That leaves $q \in \{-2, -1, 0\}$:

- $q = 0$: $p^3 = -1$, so $p = -1$.
- $q = -1$: $p^3 = -1 + 2 - 1 = 0$, so $p = 0$.
- $q = -2$: $p^3 = -8 + 8 - 1 = -1$, so $p = -1$.

All three give integer p, so the integer solutions are

$$(p, q) = (-1, 0), \quad (0, -1), \quad (-1, -2).$$

Checks

Substitute each answer back in. For example, $(-1)^3 = -1 = (-2)^3 + 2(-2)^2 - 1 = -8 + 8 - 1$. A computer search over all integers $|n|, |q| < 10^5$ finds exactly the two

solutions of $(*)$ and the three solutions of part (ii), and no others. You can also test the squeeze numerically. For $n = 5$, $m = \sqrt[3]{151} \approx 5.32$, which lies strictly between 5 and 6.

Common slips

- Forgetting $q = 0$ in part (ii). There, $p^3 - q^3 = -1 < 0$, so the lower bound $p > q$ fails, and the case needs checking separately.
- Writing "$m^3 > n^3$ so $m > n$" with no reason. It is true because cubing is strictly increasing (which fails for squaring with negative numbers), and that is worth saying.
- Stopping at $n < m < n + 1$ without saying why it means m is not an integer.

Beyond STEP

Equations such as $y^3 = f(x)$ asking for integer points are *Diophantine equations*. Trapping the unknown between consecutive powers is a standard first attack on them. Harder versions, such as $y^2 = x^3 + k$, lead to elliptic curves and to Baker's theory of linear forms in logarithms. These are deep areas, yet many cases still give way to elementary bounding like this. Davenport's *The Higher Arithmetic* is a good introduction to elementary number theory.

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